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Graph Analysis Project:
Study of a network formation mechanism

In this project, you will study the impact of an attribute on the formation of a real network. You will focus on one mechanism related to either assortativity or dissortativity on a node attribute (similar or dissimilar individuals tend to be connected), to the sociability or popularity of a node attribute (some individuals connect to more people or receive more connections), or to an edge attribute (individuals connected through another network tend to be connected). Remember, you are not expected to prove a mechanism, but to assess whether the observed network structure is consistent with it.

Please leave the instructions in italic. Attach your code as a RmD file.

1. Problem Statement

Describe the network you are studying. Which attribute are you interested in, and why is it interesting or relevant to study its effect on the formation of ties in this network? (max 300 words)

This project studies the San Antonio Spurs' assist network during the 2024-25 NBA regular season. The NBA is a professional basketball league, and the San Antonio Spurs are one of its teams. In basketball, an assist is a pass that directly helps a teammate score. In our network, each player is a node, and each directed edge shows assists from one player to another. The network is also weighted, because some pairs of players exchange many more assists than others.

The node attribute we study is player popularity, measured with the number of Instagram followers. We use this as a simple indicator of media visibility. This is interesting because some players are much more famous than others, like Victor Wembanyama for example. We want to know if this difference in popularity is related to the way connections are formed inside the team.

More precisely, we ask whether more popular players are more likely to receive assist ties from their teammates. In other words, do players tend to pass more often to teammates who are more visible and famous? This matches a popularity mechanism in graph formation: some individuals receive more connections than others because of one of their attributes.

We do not try to prove that Instagram popularity causes assists. Instead, we test whether the observed network is consistent with this mechanism. The Spurs are a good case because the team includes both very famous players and less visible players, so popularity varies a lot inside the same network.

2. Expectations

Formulate expectations about the causes and consequences of this mechanism: Why do you expect this mechanism to occur? What network characteristic(s) do you expect as an outcome of this mechanism? (max 300 words)

We expect player popularity to matter in the Spurs' assist network because popularity often reflects visibility, status, and importance inside and outside the team. More popular players are usually more central in the team's image, and they may also be seen as key offensive options. For this reason, teammates may pass to them more often, especially when they are expected to finish actions or create efficient scoring opportunities.

In our case, Instagram followers are used as a simple proxy for popularity. We know this is not a perfect basketball variable, but it can still capture differences in fame and media attention between players. The Spurs are a good case for this because popularity is very unequal across players, from globally famous players such as Victor Wembanyama to much less visible rotation players.

If this popularity mechanism is present, we expect more popular players to be more likely to receive assist ties. In network terms, we therefore expect a positive relationship between Instagram followers and the probability of receiving an incoming assist tie.

These expectations match the idea of a popularity mechanism in graph formation: some individuals receive more connections because of one of their attributes.

3. Research design

Explain how to define a QAP test to find some evidence for the mechanism you are studying. Clearly define your statistics and null model. (max 300 words)

To test our mechanism, we define a QAP regression where the dependent variable is the Spurs' assist network. We use a binary adjacency matrix X, where $\mathrm{X_{ij}} = 1$ if player i gave at least one assist to player j during the 2024–25 regular season, and $\mathrm{X_{ij}} = 0$ otherwise. The explanatory variable is player popularity, measured with Instagram followers. We then build a dyadic matrix Zsuch that $\mathrm{Z_{ij}}$ is the Instagram popularity of the receiving player j. This allows us to test whether assist ties are more likely to go to more popular players.

Our statistic of interest is the coefficient $\theta_1$of the popularity matrix in a logistic QAP regression:

$$\mathrm{logit}(\mathrm{P}(\mathrm{X_{ij}} = 1)) = \theta_0 + \theta_1 \mathrm{Z_{ij}} + \varepsilon_{\mathrm{ij}}$$

The null hypothesis is that there is no association between popularity and assist ties, so $\theta_1 = 0$. The alternative hypothesis is that more popular players are more likely to receive an assist tie so $\theta_1 > 0$.

As a null model, we use QAP permutations. We repeatedly permute the node labels in the network matrix while keeping the overall network structure unchanged. For each permutation, we estimate the regression again and compare the observed coefficient to the distribution of permuted coefficients. This gives us a empirical p-value.

Explain how you can descriptively analyze the relationship between communities in the graph and the attribute you study. Select and justify one or more statistics. (max 300 words)

To descriptively analyze the relationship between communities and our attribute, we first transform the directed assist network into an undirected graph using the collapse option. We do this because even if an assist goes only in one direction, it still shows an offensive interaction between two players. We then identify communities with the Louvain algorithm. This method is appropriate because it is based on modularity, which helps detect groups of players that are more connected to each other.

After identifying the communities, we compare them using Instagram popularity as an external attribute. Since popularity is a continuous variable, we use three simple statistics. First, we compute the mean number of followers in each community to measure the overall popularity of the group. Second, we compute the median number of followers, because it is less affected by extreme values from very famous players. Third, we compute the share of players in the top 25% of the follower distribution in each community. This helps us see whether the most popular players are concentrated in one group.

Finally, we include a graph visualization with node colors showing communities and node sizes showing Instagram followers. Together, these measures provide a simple descriptive view of the relationship between community structure and popularity.

4. Data collection

Describe your data. What are the nodes, the edges, and the attributes? How did you collect the data? Provide a graph visualization and three descriptive measures of the network that are relevant for your analysis. (max 300 words)

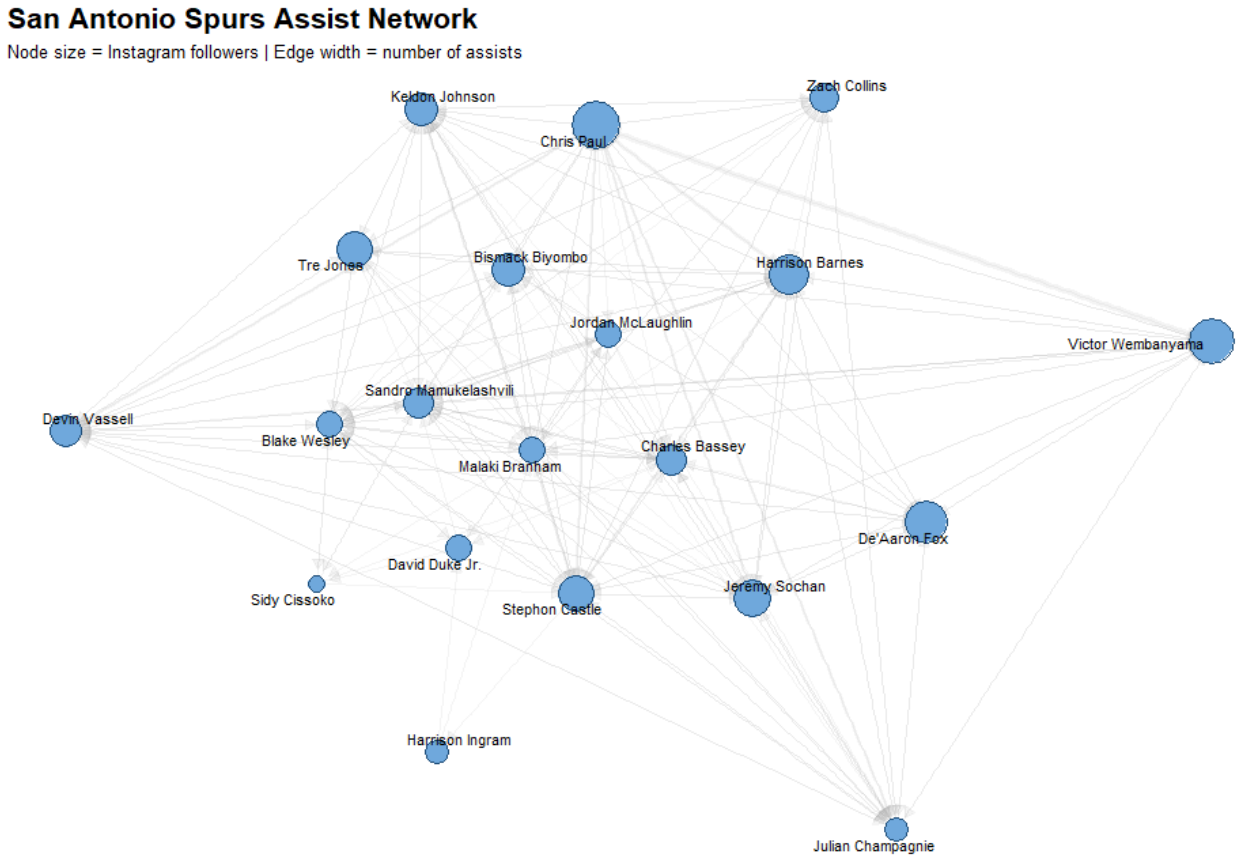
Our data describe the San Antonio Spurs' assist network during the 2024–25 NBA regular season. The nodes are the 20 players on the team. The edges are directed assist ties: an edge from player *i* to player *j* means that player *i* gave at least one assist to player *j*. The network is also weighted, because the weight of each edge shows the number of assists between two players. The main attribute in our analysis is the number of followers on Instagram for each player, which we use as a measure of popularity.

We collected the data from public online sources. We used PBP Stats to collect assist interactions between Spurs players during the regular season. Then, we collected each player's follower count from public Instagram profiles. After that, we matched the follower counts to the players and added this information as a node attribute in the graph.

The graph contains 20 nodes and 226 directed edges. The visualization shows a dense core of players who exchange many assists, and a few players who are more peripheral. In the figure, larger nodes represent players with more followers on Instagram, and thicker edges represent a higher number of assists.

Three descriptive measures are useful for our analysis. First, density is 0.595, meaning that 59.5% of all possible directed assist ties are present in the network. Second, reciprocity is 0.867, meaning that most observed assist ties belong to mutual pairs of players assisting each other. Third, transitivity is 0.800,

meaning that assist ties often form clustered local structures. Together, these measures show that the Spurs' assist network is dense, highly reciprocal, and strongly clustered.



5. Exploration and Analysis

Perform the QAP test previously described. Report the test statistics and empirical p-value. (max 200 words)

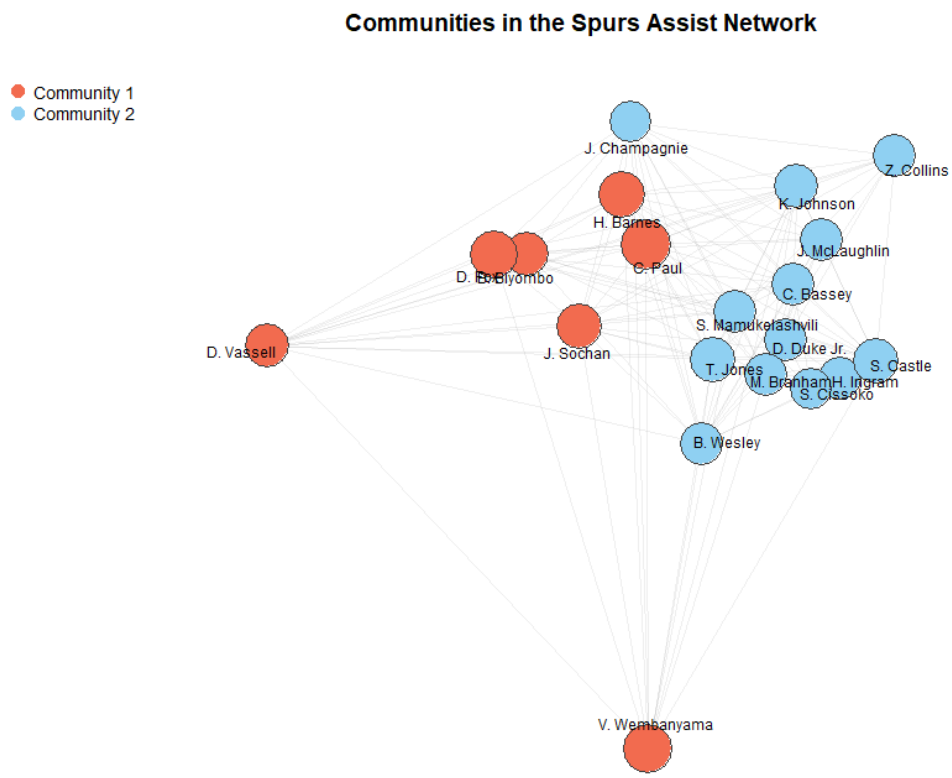
We performed the logistic QAP test described in the research design. The dependent variable is the binary assist matrix, where a value of 1 means that player i gave at least one assist to player j. The independent variable is the Instagram follower count of the receiving player. The estimated coefficient for popularity is positive but extremely small ($\beta = 4.43 \times 10^{-8}$). This small size is expected because Instagram followers are measured in raw counts. The effect is in the expected direction, which means that assist ties are slightly more likely to point toward more popular players. However, the result is not statistically significant. The empirical QAP p-value is 0.374 for the one-sided test and 0.677 for the two-sided test. Overall, we do not find strong evidence that Instagram popularity increases the probability of receiving an assist tie in the Spurs network. The result goes in the expected direction, but the effect remains too weak to support a clear popularity mechanism.

Identify communities in your graph (justify your choice of algorithm). Describe the communities you found and provide a graph visualization of the communities. (max 300 words)

To identify communities in the graph, we first transform the directed assist network into an undirected graph using the collapse option. We do this because even if an assist is observed in only one direction, it still represents an offensive interaction between two players. We then apply the Louvain algorithm. This is a good choice because it is a modularity-based method that helps identify groups of players who are more connected to each other.

The Louvain algorithm identifies two communities in the Spurs assist network. The first community contains 7 players: Chris Paul, Bismack Biyombo, Harrison Barnes, De'Aaron Fox, Devin Vassell, Jeremy Sochan, and Victor Wembanyama. The second community contains the other 13 players. Community 1 appears to include several of the team's most visible and central offensive players, while Community 2 is larger and more peripheral. However, the modularity score is low (Q = 0.096), which means that the separation between the two communities is weak. This is consistent with the fact that the network is dense and highly connected.

The following figure shows the two communities detected in the network. Node colors represent community membership.



Compute the statistic(s) proposed in section 3. Does the distribution of individuals in the communities match your expectations regarding the mechanism you are studying? Explain why or why not. (200 words)

Table 1: Instagram popularity by community

Community	Players	Mean followers	Median followers	Top 25% share
Community 1	7	2,920,000	973,000	71.4%
Community 2	13	108,769	63,000	0%

The statistics show a clear descriptive difference in Instagram popularity across the two communities. Community 1 has a much higher mean number of followers (2,920,000 vs. 108,769), a much higher median (973,000 vs. 63,000), and a much larger share of players in the top 25% of the follower distribution (71.4% vs. 0%). This means that the most popular players are concentrated in Community 1, while Community 2 contains less visible players on average. This pattern is consistent with our expectations regarding a popularity mechanism. However, this result remains descriptive, and it should be interpreted with caution because the modularity score is low, which suggests that the communities are not strongly separated.

This pattern matches our expectations regarding a popularity mechanism, because the community that looks like the main offensive core also contains the most popular players. However, this result remains descriptive. It shows that popularity is unevenly distributed across communities, but it does not prove that popularity directly causes assist ties.

6. Additional analysis

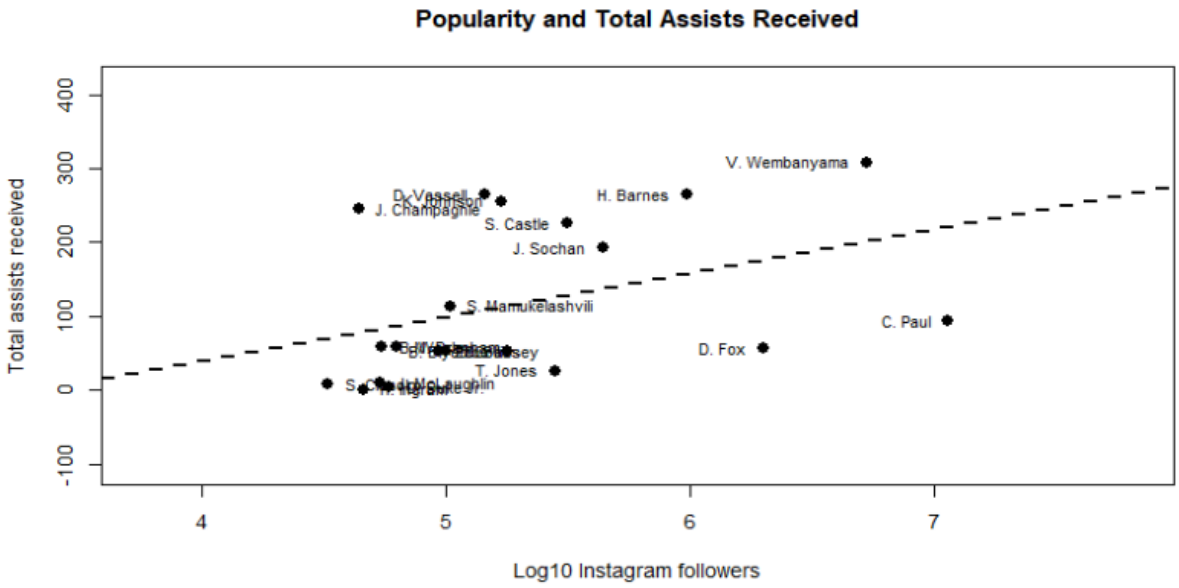
Choose an additional analysis that would deepen or challenge your previous findings. You may perform similar analyses on other graphs, run simulations, check what happens if you add nodes, remove edges, … Explain your rationale, describe your method, and interpret your results. (500 words)

As an additional analysis, we study whether Instagram popularity is related not only to the existence of assist ties, but also to the total number of assists received by each player. This is useful because our main QAP test used a binary network, so it only showed whether a tie existed or not. Here, we look at the intensity of the relationship.

We create a player-level dataset with each player's number of Instagram followers, binary in-degree, and total assists received. We then compute Spearman correlations between popularity and these two measures. The correlation between followers and binary in-degree is positive but not significant ($\rho = 0.271$, $p = 0.248$). This means that more popular players do not clearly receive assists from a larger number of different teammates. However, the correlation between followers and total assists received is stronger and significant ($\rho = 0.529$, $p = 0.017$). A permutation test confirms this result, with a one-sided empirical p-value of 0.010 and a two-sided empirical p-value of 0.016.

These results help us better understand our main findings. The QAP test gave weak evidence for a popularity mechanism in the binary network. However, this additional analysis shows that popularity is more clearly related to the total number of assists received. In other words, popular players do not necessarily receive assists from a larger number of different teammates, but they tend to receive more

assists overall. This additional analysis complements rather than replaces the QAP result, because it is conducted at the player level and focuses on weighted offensive involvement rather than on the simple existence of dyadic ties.



7. Interpretation and conclusions

Summarize what you learned from your analysis. Were your results all in line with your expectations? (max 200 words)

Overall, our analysis shows that Instagram popularity is related to the Spurs' assist network, but not in a completely clear way. The QAP test gave a positive coefficient, which goes in the expected direction, but the effect was very small and not statistically significant. This means that we do not find strong evidence that more popular players are more likely to receive an assist tie in the binary network.

However, the community analysis and the additional analysis give a more nuanced result. The community analysis shows that the most popular players are concentrated in one community, although the low modularity score suggests that this separation remains weak. The additional analysis also shows a significant positive relationship between Instagram followers and the total number of assists received. This suggests that popularity is more strongly related to the intensity of offensive involvement than to the simple existence of ties.

Overall, we do not find strong evidence for a popularity mechanism in binary tie formation. However, complementary analyses suggest that popularity may be more related to the intensity of assist reception than to the simple existence of ties.

Discuss the limitations of this study and identify possible ways to improve or enrich your study. Could this type of analysis be used to provide insight into or answer real-life problems? (max 300 words)

This study has several limitations. First, we only study one NBA team during one regular season. Because of that, our results are very specific to the Spurs and cannot be generalized too much. Another team could have a different playing style, a different group of star players, or a different balance between popularity and on-court performance. It is also possible that the same team would show different results in another season.

Second, Instagram followers are not a perfect measure of popularity. They are useful because they are easy to observe and compare, but they only capture one side of popularity. A player can have many followers because of media exposure, personality, or past success, and not only because of his current role in the team. In the same way, some important players may have fewer followers than expected. So this variable is interesting, but it is still imperfect.

Third, our main QAP analysis is based on a binary network. This means that we only study whether an assist tie exists or not. We do not fully capture how often players interact. This is important because our additional analysis suggests that popularity may be linked more strongly to the total number of assists received than to the simple existence of a tie. In that sense, the binary network may simplify the reality too much.

This study could be improved in several ways. One possible improvement would be to compare several teams or several seasons. Another would be to include more player attributes, such as minutes played, salary, points per game, or position. This would help us see whether popularity still matters when other factors are taken into account. It would also be useful to work more with weighted networks, since assist intensity seems important in our results.

This type of analysis can be useful for real-life problems. In sports, it can help better understand team coordination, offensive structure, and the role of key players. More generally, similar methods can be used in companies, schools, or online platforms to study how status, visibility, or reputation affect interactions between people